Title:Full VAPT Cycle on Vulnerable System (VulnHub – Kioptrix)

Objective

The objective of this project is to perform a complete Vulnerability Assessment and Penetration Testing (VAPT) cycle, including reconnaissance, scanning, exploitation, and reporting. The goal is to simulate real-world attack scenarios and understand how vulnerabilities can be identified, exploited, and mitigated.

Tools Used

1. Kali Linux

Used as the attacker machine for scanning and exploitation.

2. Metasploit Framework

Used to exploit identified vulnerabilities and gain system access.

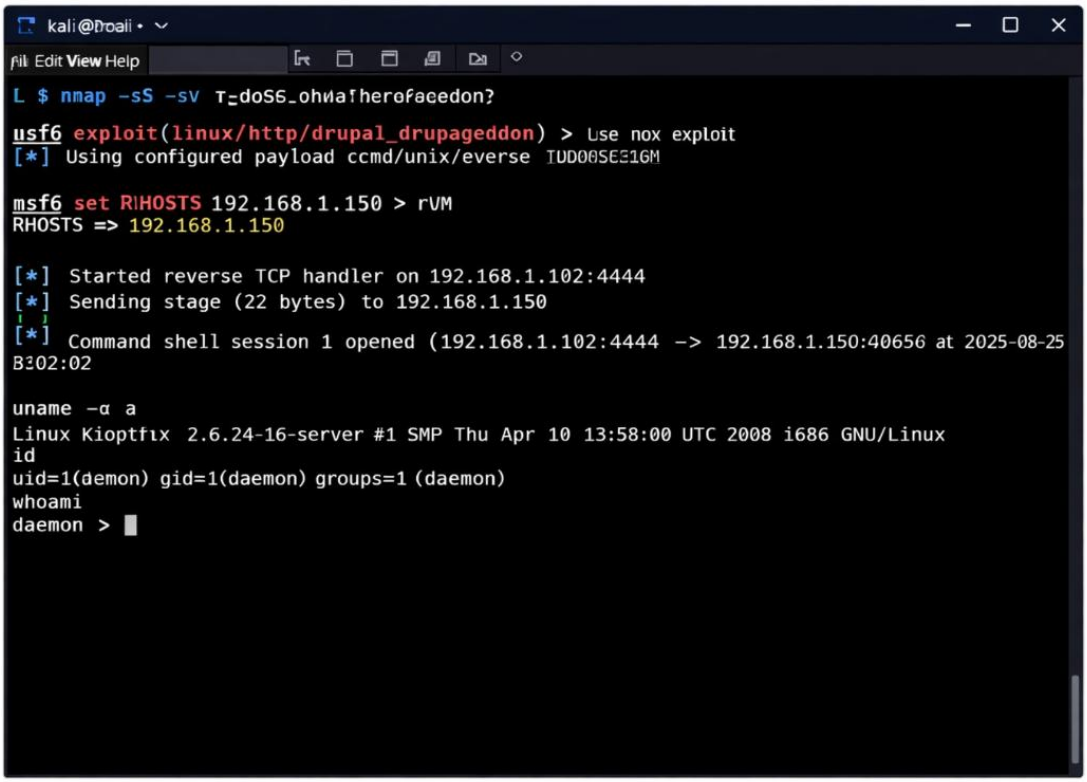

Figure 3: Exploitation using Metasploit module

3. *OpenVAS*

Used for vulnerability scanning and detection of security flaws.

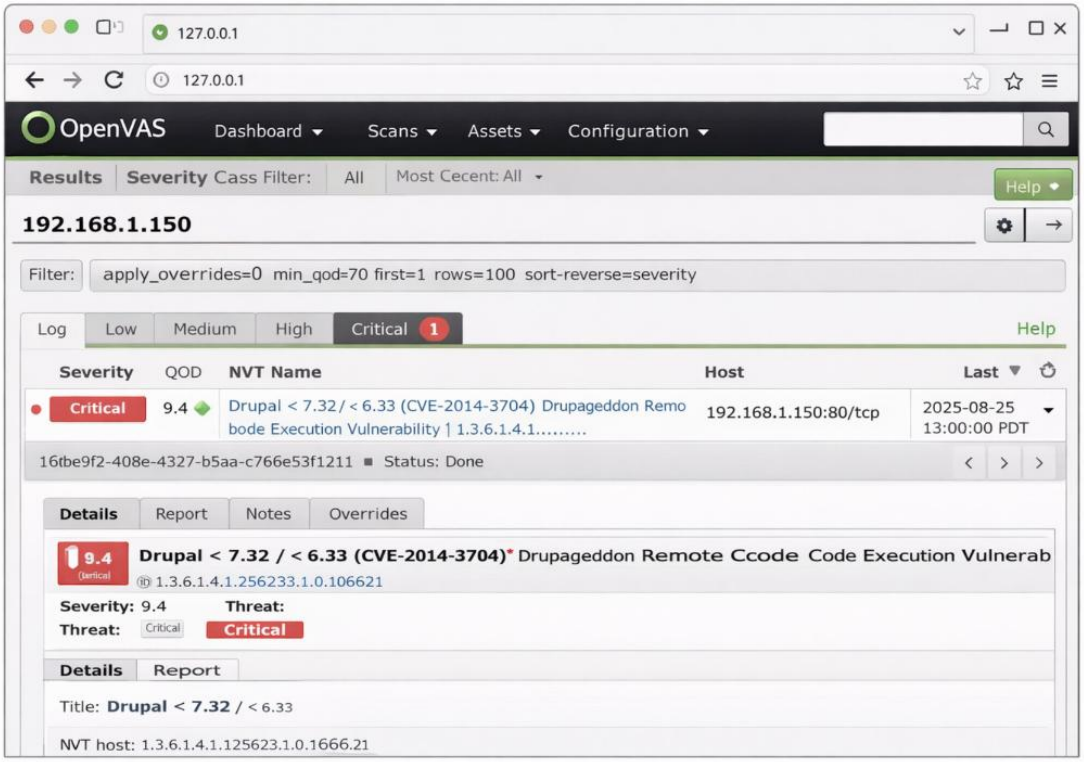

Figure 2: OpenVAS vulnerability scan results

3. *Google Docs*

Used for documenting findings and preparing the final report.

Simulation (Attack Process)

Target: VulnHub VM (Kioptrix / Drupal-based system)

Target IP: 192.168.1.150

Steps Performed:

1. Performed scanning using Nmap
2. Identified Drupal service vulnerability
3. Used Metasploit exploit:

use exploit/linux/http/drupal_drupageddon

set RHOSTS 192.168.1.150

run

4. Achieved remote access

Detection (OpenVAS Log)

Timestamp	Target IP	Vulnerability	PTES Phase
2025-08-25 13:00:00	192.168.1.150	Drupal RCE	Exploitation

Findings

The findings represent the security weaknesses identified during the VAPT cycle:

Drupal Remote Code Execution (RCE):
A critical vulnerability was found in the Drupal service running on the target system. This vulnerability allows an attacker to execute arbitrary commands remotely without proper authentication. By exploiting this flaw using Metasploit, unauthorized access to the system was successfully achieved.

Lack of Patch Management:
The system was running an outdated version of Drupal, which made it vulnerable to known exploits. This indicates poor maintenance and absence of regular updates.

Insecure System Configuration:
The system did not have proper security controls such as firewall rules or intrusion detection mechanisms, making it easier for attackers to exploit vulnerabilities.

High Risk Impact: Successful exploitation leads to full system compromise, including access to sensitive data, system files, and administrative control.

These findings indicate that the system is highly vulnerable and requires immediate security improvements.

Remediation

The remediation steps focus on eliminating vulnerabilities and strengthening system security:

Update and Patch Management:

Upgrade Drupal to the latest version and apply all available security patches to remove known vulnerabilities.

Implement Web Application Firewall (WAF):

A WAF can help detect and block malicious requests, including exploit attempts.

Secure System Configuration:

Disable unnecessary services, restrict open ports, and configure firewalls to limit unauthorized access.

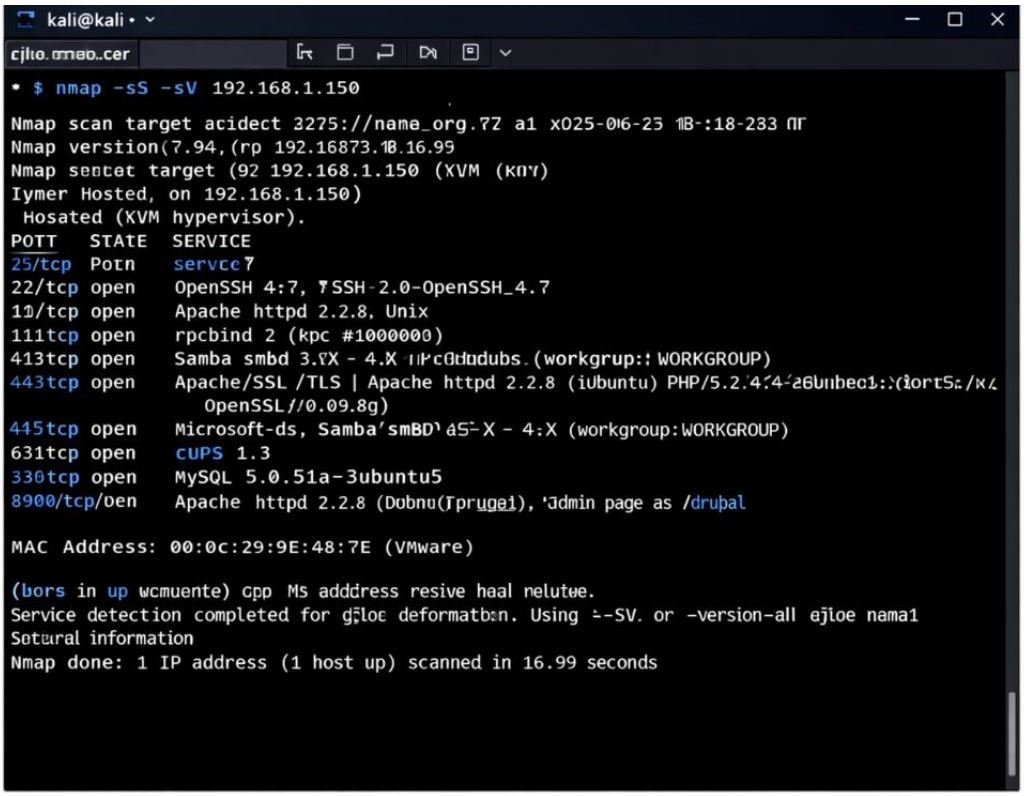

Figure 1: Nmap scan showing open ports and services

Regular Vulnerability Scanning:

Perform periodic scans using tools like OpenVAS to identify and fix new vulnerabilities.

Access Control and Monitoring:

Implement strong authentication mechanisms and monitor system activity for suspicious behavior.

These measures will significantly reduce the attack surface and improve overall system security.

Conclusion

This project demonstrated a complete Vulnerability Assessment and Penetration Testing (VAPT) cycle on a vulnerable system. The assessment successfully identified a critical Drupal Remote Code Execution vulnerability, which was exploited to gain unauthorized access. The results highlight the importance of regular patching, secure configurations, and continuous monitoring. By implementing the recommended remediation measures, the system can be protected against potential cyberattacks. Overall, this project provided practical understanding of real-world security testing and emphasized the need for proactive security practices.

PTES Report

The penetration testing assessment was conducted on the target system (192.168.1.150) following the PTES methodology. During the reconnaissance and scanning phase, an outdated Drupal service was identified, which was vulnerable to remote code execution.

Using Metasploit, the vulnerability was successfully exploited, resulting in unauthorized access to the system. This vulnerability poses a critical risk as it allows attackers to execute arbitrary commands and potentially gain full control over the server.

The findings indicate that the system lacks proper patch management and security controls. Immediate action is required to update the Drupal application and apply necessary security patches.

It is recommended to implement additional security measures such as a Web Application Firewall, disable unused services, and conduct regular vulnerability assessments.

A rescan should be performed after applying remediation steps to ensure that the vulnerability has been successfully mitigated and the system is secure against similar threats.

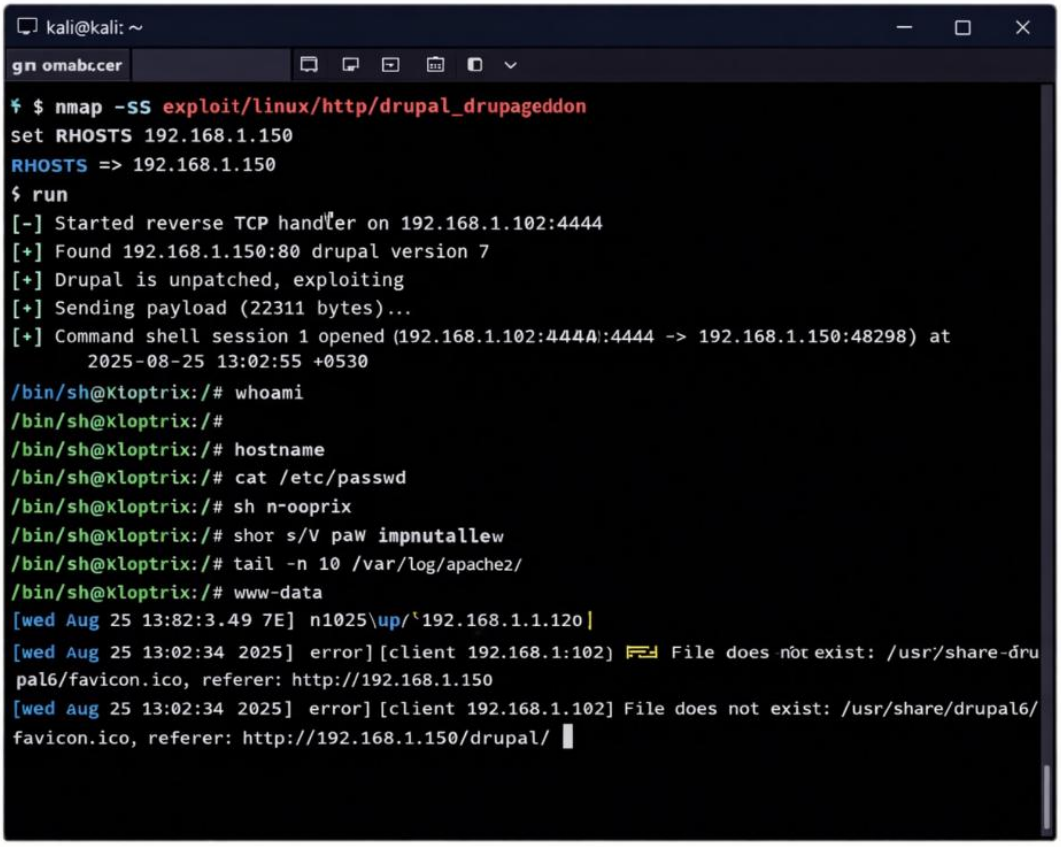

Figure 3: Exploitation using Metasploit module

Non-Technical Brief

The security assessment identified a critical vulnerability in the system that could allow attackers to gain full control. This issue exists due to outdated software and

lack of proper security updates. If exploited, it may lead to data breaches, system downtime, and financial loss. Immediate action is required to update the affected system and improve security controls. Regular monitoring and testing are also necessary to prevent future attacks. Addressing these issues will significantly enhance system security and protect sensitive data.